

LINEAR DATA STRUCTURES BASICS

Lists

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Course Outline



ADT Concept and Definition

Definition

An **Abstract Data Type** (ADT) is a mathematical model that defines data and operations without specifying how they are implemented.

Key Characteristics of ADTs:

- **Data Abstraction:** Focus on what operations do, not how they are implemented.
 - **Encapsulation:** Hide implementation details.
 - **Interface:** Define clear methods for interaction.
 - **Implementation:** Implementation can change without affecting users.
- Native / primitive** { integer, double, float, boolean, char
- Object** → **Class**
- no pointers**
- Car** { brand, color, motor, max speed }
- VideoGame**



ADT Concept and Definition

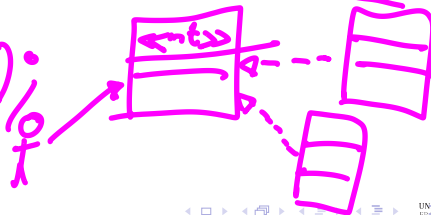
Definition

An **Abstract Data Type (ADT)** is a mathematical model that defines data and operations without specifying how they are implemented.

Key Characteristics of ADTs:

- **Data Abstraction:** Focus on what operations do, not how
- **Encapsulation:** Hide implementation details, GET/SET
- **Interface:** Define clear methods for interaction
- **Independence:** Implementation can change without affecting users

Type Name (param):



ADT Operations

Example

List ADT Operations:

- Add element to list
- Remove element from list
- Find element in list
- Get size of list
- Check if list is empty

→ Create
→ Retrieve / lead
→ Update
→ Delete

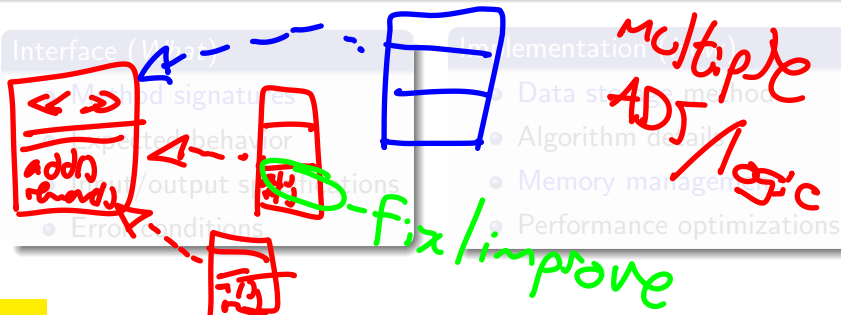
→ Encapsulation



Interface vs. Implementation Separation I

Why Separate *Interface* from *Implementation*?

- **Flexibility:** Can change how things work internally
- **Maintainability:** Easier to fix and improve code
- **Reusability:** Same interface, different implementations
- **Testing:** Can test interface independently



Interface vs. Implementation Separation I

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- **Reusability:** Same interface, *different* implementations
- **Testing:** Can test interface *independently*

Interface (*What*)

- Method signatures
- Expected behavior
- Input/output specifications
- Error conditions

Implementation (*How*)

- *type* *range (params)*
• *output* *input*
• Algorithm details
- Memory management
- Performance optimizations



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Interface (*What*)

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Implementation (*How*)

- Data storage method
- Algorithm details
- Memory management
- Performance optimizations

child
classes

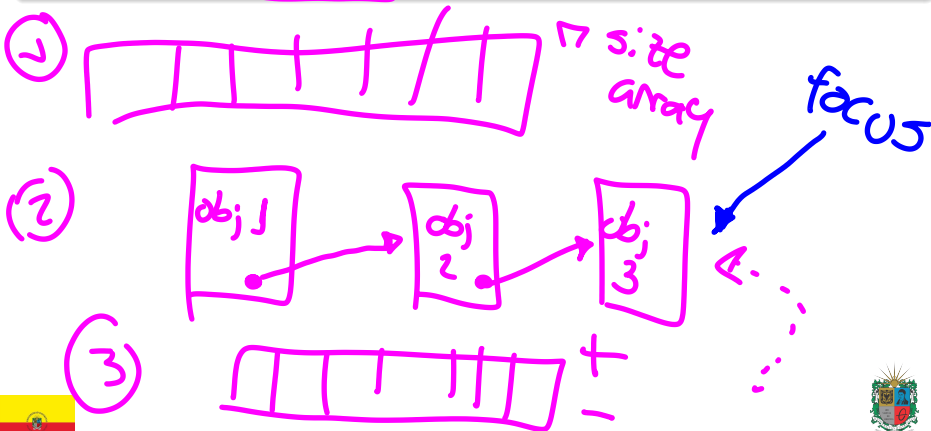


Interface vs. Implementation Separation II

Example

List Interface: `add(item)`, `remove(item)`, `size()`

Implementations: Array-based, Linked-list-based, Dynamic array



Study Case: Student Enrollment System Using Lists I

Problem

A university needs a system to manage student enrollment in courses. Students can be added, removed, and searched in course rosters.

Example

System Requirements:

- Add student to course
- Remove student from course
- Check if student is enrolled
- List all enrolled students
- Get enrollment count
- Transfer student between courses

Study Case: Student Enrollment System Using Lists II

EnrollmentSystem
+ addStudent(courseId, student) + removeStudent(courseId, student) + isEnrolled(courseId, student) + getEnrollmentCount(courseId) + listStudents(courseId)

A possible solution:

class
+
methods
↓
ADT

Benefits of Using Lists

- Dynamic size as enrollment changes
- Easy to add/remove students
- Can maintain enrollment order
- Efficient searching capabilities



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retrieve search / most expensive



Static vs. Dynamic Lists I

Static Lists

Characteristics:

- Fixed size at creation
- Uses arrays as base structure
- Memory allocated at compile time
- Fast access by index

Dynamic Lists

Characteristics:

- Size changes during execution
- Uses pointers as base structure
- Memory allocated at runtime
- Flexible size management

When to Use Each

- Static: Known size, frequent access, simple operations
- Dynamic: Unknown size, frequent insertions/deletions



Static vs. Dynamic Lists I

Static Lists

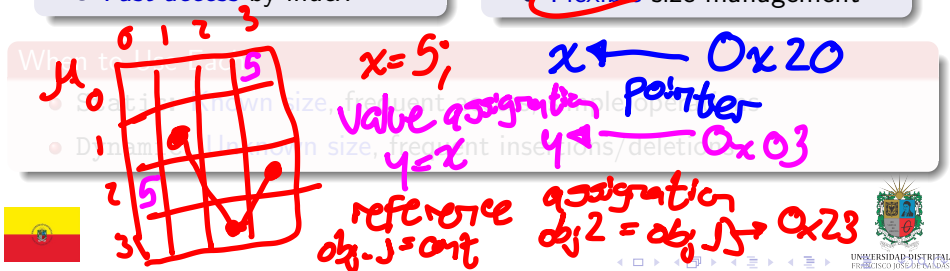
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Static vs. Dynamic Lists II

Static Lists

Advantages:

- Simple implementation
- Fast random access
- No memory overhead

Disadvantages:

- Fixed maximum size
- Memory waste if not full
- Insertion/deletion expensive

Dynamic Lists

Advantages:

- No size limitations
- Efficient memory usage
- Easy insertion/deletion

Disadvantages:

- More complex implementation
- Memory overhead for pointers
- Slower access in some cases

Null



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HEAD-SIZE



List Operations I: Insert, Search

Core List Operations

All list implementations must support **fundamental operations** for data manipulation.

Insert Operations

- Insert at **beginning**
- Insert at **end**
- Insert at **specific position**
- Insert in **sorted order**

Search Operations

- Find element by **value**
- Find element at **position**
- Check if element **exists**
- Count **occurrences**



List Operations I: Insert, Search

Core List Operations

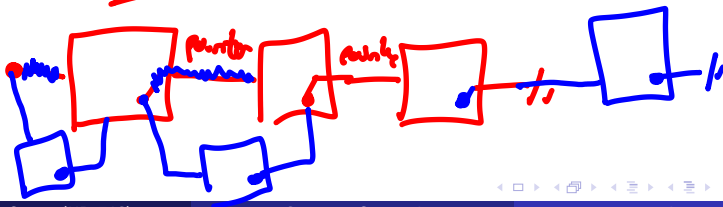
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List Operations II: Delete, Update

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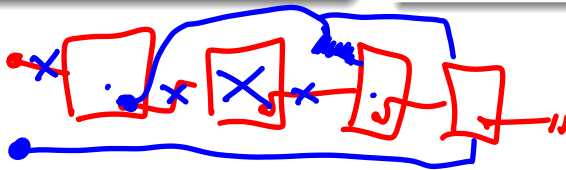
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Delete Operations

- Delete by value
- Delete at position
- Delete first occurrence
- Delete all occurrences

Update Operations

- Update element at position
- Update first occurrence
- Update all occurrences
- Swap elements



List Operations II: Delete, Update

Core List Operations

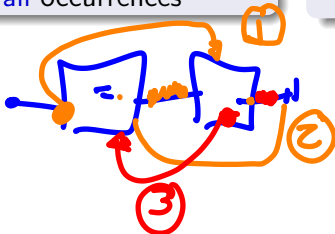
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- Delete **first** occurrence
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Update Operations

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- Update **all** occurrences
- **Swap** elements



Study Case: Shopping Cart Implementation with Array-based List I

Problem

Implement a shopping cart system where customers can add items, remove items, update quantities, and view their cart contents.

↓
dynamical

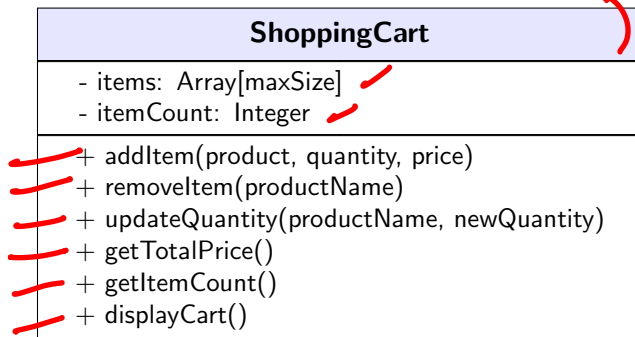
ShoppingCart	
- items: Array[maxSize]	C 2 4 0
- itemCount: Integer	
+ addItem(product, quantity, price)	
+ removeItem(productName)	
+ updateQuantity(productName, newQuantity)	
+ getTotalPrice()	
+ getItemCount()	
+ displayCart()	



Study Case: Shopping Cart Implementation with Array-based List I

Problem

Implement a **shopping cart** system where customers can **add** items, **remove** items, **update quantities**, and view their cart contents.



Study Case: Shopping Cart Analysis with *Array-based List II*

Array Implementation Benefits

- Fast access to any item by index
- Simple memory layout
- Good for small, bounded carts
- Easy to implement sorting (by price, name)

Considerations

- Need to handle cart size limit
- Shifting elements when inserting
- Memory allocated upfront
- Good for typical shopping scenarios (few items)

typical fields



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Dynamic Memory Allocation I

Definition

Dynamic memory allocation allows programs to **request memory during execution**, enabling data structures to **grow and shrink** as needed.

Key Concepts

- **Runtime Allocation:** Memory requested when needed
- **Heap Memory:** Memory pool for dynamic allocation
- **Pointers:** Variables that store memory addresses
- **Memory Management:** Allocation and deallocation



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Dynamic Memory Allocation II

Advantages

- Flexible memory usage
- No size limits at compile time
- Memory used only when needed
- Can handle varying data sizes

Challenges

- Memory leaks possible
- Fragmentation issues
- Slower than static allocation
- More complex programming

Best Practices

- Always free allocated memory
- Check for allocation failures
- Avoid memory leaks
- Use appropriate allocation strategies

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Pointer-based List Structures I

Definition

Pointer-based lists use **dynamic memory allocation** and **pointers** to create flexible data structures where elements can be stored **anywhere in memory**.

Node Structure

Each element (**node**) contains:

- **Data:** The actual **information** stored
- **Pointer:** Reference to the next element
- **Optional:** Additional pointers for **complex structures**



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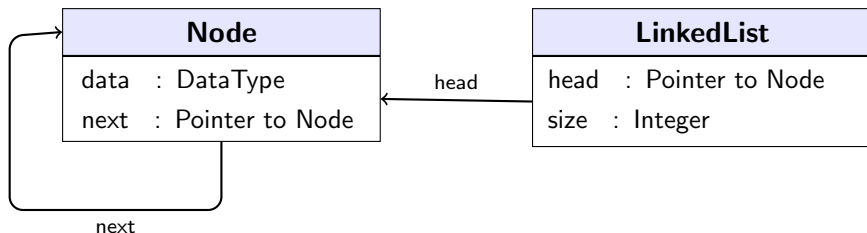
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Pointer-based List Structures II



Pointer-based List Structures III

Advantages

- Dynamic size
- Efficient insertion/deletion
- Memory allocated as needed
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Disadvantages

- No random access
- Extra memory for pointers
- Sequential access only
- More complex implementation



Pointer-based List Structures III

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- Memory allocated as needed
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Disadvantages

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Study Case: Music Playlist Management with *Pointer-based List I*

Problem

Create a **music playlist** system where songs can be easily **added**, **removed**, and **reordered**. The playlist should handle an **unknown number** of songs efficiently.

Structure Song

title: String

artist: String

duration: Integer

next: Pointer to Song

end structure

Class Playlist

head: Pointer to Song

current: Pointer to Song

totalSongs: Integer



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Study Case: Music Playlist Management with *Pointer-based List II*

Playlist Operations

- **Add Song:** Insert at **end** or specific position
- **Remove Song:** Delete by **title** or position
- **Play Next:** Move to **next song** in sequence
- **Shuffle:** Rearrange song order randomly
- **Search:** Find song by **title** or **artist**

Why *Linked List* Works Well

- Unknown number of songs
- Frequent additions/removals
- Sequential playback matches structure
- Easy to implement shuffle and repeat

Study Case: Music Playlist Management with *Pointer-based List II*

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Thanks!

Questions?



Repo: <https://github.com/EngAndres/ud-public/tree/main/courses/computer-sciences-i>

